

LISTA de G.A. gabriela Vinha Feitosa

1- a) $A+2B$ $\begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix} + \begin{pmatrix} 0 & 10 \\ 6 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 10 \\ 2 & 3 \end{pmatrix}$$

b) $AB - BA$ $\begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix} \cdot \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix} - \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix}$

$$\begin{pmatrix} 1 \cdot 0 + 0 \cdot 3 & 1 \cdot 5 + 0 \cdot (-2) \\ 2 \cdot 0 + 7 \cdot 3 & 2 \cdot 5 + 7 \cdot (-2) \end{pmatrix} - \begin{pmatrix} 0 \cdot 1 + 5 \cdot 2 & 0 \cdot 0 + 5 \cdot 7 \\ 3 \cdot 1 + (-2) \cdot 2 & 3 \cdot 0 + (-2) \cdot 7 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 5 \\ 21 & -4 \end{pmatrix} - \begin{pmatrix} 10 & 35 \\ -1 & -14 \end{pmatrix}$$

$$\therefore AB - BA = \begin{pmatrix} -10 & -30 \\ 22 & 10 \end{pmatrix}$$

c) $2C - D$ $2 \cdot \begin{pmatrix} -2 & 3 & -7 \\ 7 & -3 & -2 \end{pmatrix} \cdot \begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} \Rightarrow \begin{pmatrix} -4 & 6 & -14 \\ 14 & -6 & -4 \end{pmatrix} \cdot \begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} =$

$$= \begin{pmatrix} -4 \cdot (-3) + 6 \cdot 1 + (-14) \cdot (-2) & -4 \cdot 2 + 6 \cdot 1 + (-14) \cdot 0 & -4 \cdot 0 + 6 \cdot 4 + (-14) \cdot 2 \\ 14 \cdot (-3) + (-6) \cdot 1 + (-4) \cdot (-2) & 14 \cdot 2 + (-6) \cdot 1 + (-4) \cdot 0 & 14 \cdot 0 + (-6) \cdot 4 + (-4) \cdot 2 \end{pmatrix} = \begin{pmatrix} 46 & -2 & -4 \\ -40 & 22 & -32 \end{pmatrix}$$

d) $2D^t - 3E^t$

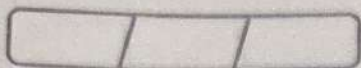
$$2 \cdot \begin{pmatrix} -3 & 1 & -2 \\ 2 & 1 & 0 \\ 0 & 4 & 2 \end{pmatrix} - 3 \cdot \begin{pmatrix} 2 & -1 & -6 \\ 4 & 0 & 0 \\ -3 & -4 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -6 & 2 & -4 \\ 4 & 2 & 0 \\ 0 & 8 & 4 \end{pmatrix} - \begin{pmatrix} 6 & -3 & -18 \\ 12 & 0 & 0 \\ -9 & -12 & -3 \end{pmatrix}$$

$$\therefore 2D^t - 3E^t = \begin{pmatrix} -12 & 5 & 14 \\ -8 & 2 & 0 \\ 9 & 20 & 7 \end{pmatrix}$$



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e) $D^2 + DE$

$$\begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix}^2 + \begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} + \begin{pmatrix} (-3) \cdot 2 + 2 \cdot (-1) + 0 \cdot (-6) & (-3) \cdot 4 + 2 \cdot 0 + 0 \cdot 0 & -3 \cdot (-3) + 2 \cdot (4) + 0 \cdot (-1) \\ 1 \cdot 2 + 1 \cdot (-1) + 4 \cdot (-6) & 1 \cdot 4 + 1 \cdot 0 + 4 \cdot 0 & 1 \cdot (-3) + 1 \cdot (-4) + 4 \cdot (-1) \\ (-2) \cdot 2 + 0 \cdot (-1) + 2 \cdot (-6) & -2 \cdot 4 + 0 \cdot 0 + 2 \cdot 0 & -2 \cdot (-3) + 0 \cdot (4) + 2 \cdot (-1) \end{pmatrix}$$

$$\begin{pmatrix} (-3) \cdot (-3) + 2 \cdot 1 + 0 \cdot (-2) & (-3) \cdot 2 + 2 \cdot 1 + 0 \cdot 0 & (-3) \cdot 0 + 2 \cdot 4 + 0 \cdot 2 \\ 1 \cdot (-3) + 1 \cdot 1 + 4 \cdot (-2) & 1 \cdot 2 + 1 \cdot 1 + 4 \cdot 4 & 1 \cdot 0 + 1 \cdot 4 + 4 \cdot 2 \\ (-2) \cdot (-3) + 0 \cdot 1 + 2 \cdot (-2) & (-2) \cdot 2 + 0 \cdot 1 + 2 \cdot 0 & (-2) \cdot 0 + 0 \cdot 4 + 2 \cdot 2 \end{pmatrix} + \begin{pmatrix} -8 & -12 & 1 \\ -23 & 4 & -11 \\ -16 & -8 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 11 & 4 & 8 \\ 10 & 19 & 12 \\ 2 & -4 & 4 \end{pmatrix} + \begin{pmatrix} -8 & -12 & 1 \\ -23 & 4 & -11 \\ -16 & -8 & 4 \end{pmatrix}$$

$$\therefore D^2 + DE = \begin{pmatrix} 3 & -8 & 9 \\ -13 & 23 & 1 \\ -14 & -12 & 8 \end{pmatrix}$$

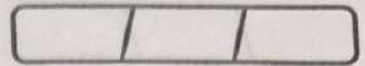
f) $C^t \cdot A$

$$\begin{pmatrix} -2 & 7 \\ 3 & -3 \\ -7 & -2 \end{pmatrix}_{3 \times 2} \cdot \begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix}_{2 \times 2}$$

$$\begin{pmatrix} (-2) \cdot 1 + 7 \cdot 2 & -2 \cdot 0 + 7 \cdot 7 \\ 3 \cdot 1 + (-3) \cdot 2 & 3 \cdot 0 + (-3) \cdot 7 \\ -7 \cdot 1 + (-2) \cdot 2 & -7 \cdot 0 + (-2) \cdot 7 \end{pmatrix}$$

$$\therefore C^t \cdot A = \begin{pmatrix} 12 & 49 \\ -3 & -21 \\ -11 & -14 \end{pmatrix}$$





$$g) E-AC \quad \begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix} - \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} -2 & 3 & -7 \\ 7 & -3 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix} - \begin{pmatrix} 0 \cdot (-2) + 5 \cdot 7 & 0 \cdot 3 + 5 \cdot (-3) & 0 \cdot (-7) + 5 \cdot (-2) \\ 3 \cdot (-2) + 7 \cdot (-2) & 3 \cdot 3 + (-2) \cdot (-3) & 3 \cdot (-7) + (-2) \cdot (-2) \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix} - \begin{pmatrix} 35 & -15 & -10 \\ -20 & 15 & -17 \end{pmatrix}$$

$\therefore E-AC$ é indefinido

$$h) f^t \cdot E \quad (1 \ -2 \ 0) \cdot \begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix}$$

$$(1 \cdot 2 + (-2) \cdot (-1) + 0 \cdot (-6) \quad 1 \cdot 4 + (-2) \cdot 0 + 0 \cdot 0 \quad 1 \cdot (-3) + (-2) \cdot (-4) + 0 \cdot (-1))$$

$$\therefore f^t \cdot E \quad (4 \ 4 \ 5)$$

$$i) BCF \quad \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} -2 & 3 & -7 \\ 7 & -3 & -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \cdot (-2) + 5 \cdot 7 & 0 \cdot 3 + 5 \cdot (-3) & 0 \cdot (-7) + 5 \cdot (-2) \\ 3 \cdot (-2) + (-2) \cdot 7 & 3 \cdot 3 + (-2) \cdot (-3) & 3 \cdot (-7) + (-2) \cdot (-2) \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 35 & -15 & -10 \\ -20 & 15 & -25 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 35 \cdot 1 + (-15) \cdot (-2) + (-10) \cdot 0 \\ -20 \cdot 1 + 15 \cdot (-2) + (-25) \cdot 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 65 \\ -50 \end{pmatrix}$$



2- a) $C_{2 \times 4}$ Definido

b) $C_{4 \times 2}$ Definido

c) Não Definido $2 \neq 3$

d) $C_{5 \times 3}$ Definido

e) Não Definido $4 \neq 3$

f) $C_{9 \times 9}$ Definido

g) $C_{2 \times 3}$ Definido

h) $C_{2 \times 2}$ Definido

3- a) $A = (a_{ij})_{2 \times 3}$, onde $a_{ij} = 3i - 2j$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \Rightarrow \begin{pmatrix} 3 \cdot 1 - 2 \cdot 1 & 3 \cdot 1 - 2 \cdot 2 & 3 \cdot 1 - 2 \cdot 3 \\ 3 \cdot 2 - 2 \cdot 1 & 3 \cdot 2 - 2 \cdot 2 & 3 \cdot 2 - 2 \cdot 3 \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} 1 & -1 & -3 \\ 4 & 2 & 0 \end{pmatrix}$$

b) $B = (b_{ij})_{3 \times 3}$, onde $b_{ij} = \begin{cases} 3i + j, & \text{se } i = j \\ i^2 - j, & \text{se } i \neq j \end{cases}$

$$\begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \Rightarrow \begin{pmatrix} 3 \cdot 1 + 1 & 1^2 - 2 & 1^2 - 3 \\ 2^2 - 1 & 3 \cdot 2 + 2 & 2^2 - 3 \\ 3^2 - 1 & 3^2 - 2 & 3 \cdot 3 + 3 \end{pmatrix} = \begin{pmatrix} 4 & -1 & -2 \\ 3 & 8 & 1 \\ 8 & 7 & 12 \end{pmatrix}$$

c) $C = (c_{ij})_{1 \times 4}$, onde $c_{ij} = j^2$

$$(c_{11} \ c_{12} \ c_{13} \ c_{14}) \Rightarrow (1^2 \ 2^2 \ 3^2 \ 4^2) \Rightarrow (1 \ 2 \ 3 \ 4)$$

d) $D = (d_{ij})$, onde $d_{ij} = \begin{cases} i^2 + j^2, & \text{se } i = j \\ 2ij, & \text{se } i \neq j \end{cases}$

$$\begin{pmatrix} d_{11} & d_{12} & d_{13} & d_{14} \\ d_{21} & d_{22} & d_{23} & d_{24} \\ d_{31} & d_{32} & d_{33} & d_{34} \\ d_{41} & d_{42} & d_{43} & d_{44} \end{pmatrix} \Rightarrow \begin{pmatrix} 1^2 + 1^2 & 2 \cdot 1 \cdot 2 & 2 \cdot 1 \cdot 3 & 2 \cdot 1 \cdot 4 \\ 2 \cdot 2 \cdot 1 & 2^2 + 2^2 & 2 \cdot 2 \cdot 3 & 2 \cdot 2 \cdot 4 \\ 2 \cdot 3 \cdot 1 & 2 \cdot 3 \cdot 2 & 3^2 + 3^2 & 2 \cdot 3 \cdot 4 \\ 2 \cdot 4 \cdot 1 & 2 \cdot 4 \cdot 2 & 2 \cdot 4 \cdot 3 & 4^2 + 4^2 \end{pmatrix}$$

$$\therefore D = \begin{pmatrix} 2 & 4 & 6 & 8 \\ 4 & 8 & 12 & 16 \\ 6 & 12 & 18 & 24 \\ 8 & 16 & 24 & 32 \end{pmatrix}$$



4. a) Sabendo que $[BA]_{23} = a_{13} \cdot b_{21} + a_{23} \cdot b_{22} + a_{33} \cdot b_{23}$

$$[BA]_{23} = 1 \cdot 2 + 2 \cdot (-1) + 5 \cdot 4$$

$$\therefore [BA]_{23} = 20$$

b) Sabendo que $[AB]_{23} = b_{13} \cdot a_{21} + b_{23} \cdot a_{22} + b_{33} \cdot a_{23}$

$$[AB]_{23} = (-2) \cdot 3 + (-3) \cdot 9 + 2 \cdot (-7)$$

$$\therefore [AB]_{23} = -52$$

c) Sabendo que $[B^t]_{31} = b_{31} \cdot b_{11} + b_{32} \cdot b_{21} + b_{33} \cdot b_{31}$

$$[B^t]_{31} = -3 \cdot 1 + (-1) \cdot 2 + (-7) \cdot (-3)$$

$$[B^t]_{31} = -22$$

d) $\text{tr}(A) = a_{11} + a_{22} + a_{33}$

$$\text{tr}(A) = 1 + (-3) + 5$$

$$\text{tr}(A) = 3$$

e) $\text{tr}(B^t) = [B^t]_{11} + [B^t]_{22} + [B^t]_{33}$

Sabendo que $\text{tr}(B^t) = \text{tr}(B) \Rightarrow \text{tr}(B) = B_{11} + B_{22} + B_{33}$

$$\text{tr}(B) = 1 + (-1) + (-7)$$

$$\text{tr}(B) = -7$$

f) $\text{tr}(A-B) = (A-B)_{11} + (A-B)_{22} + (A-B)_{33}$

Sabendo que $\text{tr}(A-B) = \text{tr}(A) - \text{tr}(B)$, então

$$\text{tr}(A-B) = 3 - (-7)$$

$$\text{tr}(A-B) = 10$$

g) $\text{tr}(AB) = (AB)_{11} + (AB)_{22} + (AB)_{33}$

$$\text{tr}(AB) = 4 + 9 + (-42)$$

$$\text{tr}(AB) = -29$$



$$5-a) 2X + A = 3B + C$$

$$2X + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = 3 \cdot \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

$$2X + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 12 & 9 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

$$2X = \begin{pmatrix} 6 & 5 \\ 13 & 9 \end{pmatrix} - \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix}$$

$$X = \begin{pmatrix} 7 & -2 \\ 11 & 3 \end{pmatrix} \div 2$$

$$X = \begin{pmatrix} \frac{7}{2} & -1 \\ \frac{11}{2} & \frac{3}{2} \end{pmatrix}$$

$$b) Y + A = \frac{1}{2} \cdot (B - C)^t$$

$$Y + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = \frac{1}{2} \left\{ \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} - \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \right\}^t$$

$$Y + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = \frac{1}{2} \left\{ \begin{pmatrix} 2 & -1 \\ 3 & 3 \end{pmatrix} \right\}^t$$

$$Y + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 3 \\ -1 & 3 \end{pmatrix}^t$$

$$Y = \begin{pmatrix} 1 & \frac{3}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix} - \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix}$$

$$Y = \begin{pmatrix} 2 & \frac{11}{2} \\ -\frac{5}{2} & \frac{9}{2} \end{pmatrix}$$

$$c) 3X + A = B - X$$

$$4X = B - A$$

$$X = (B - A) \div 4$$

$$X = \left\{ \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} - \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} \right\} \div 4$$

$$X = \begin{pmatrix} 3 & -6 \\ 2 & -3 \end{pmatrix} \div 4$$

$$X = \begin{pmatrix} \frac{3}{4} & -\frac{3}{2} \\ \frac{1}{2} & -\frac{3}{4} \end{pmatrix}$$



$$5) \begin{cases} X + Y = 3A \end{cases}$$

$$\begin{cases} X - Y = 2B + C \end{cases}$$

$$2X + 0Y = 3A + 2B + C$$

$$2X = 3 \cdot \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} + 2 \cdot \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

$$2X = \begin{pmatrix} -3 & 21 \\ 6 & 18 \end{pmatrix} + \begin{pmatrix} 4 & 2 \\ 8 & 6 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

$$2X = \begin{pmatrix} 1 & 25 \\ 15 & 24 \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & 25 \\ 15 & 24 \end{pmatrix} \div 2$$

$$X = \begin{pmatrix} \frac{1}{2} & \frac{25}{2} \\ \frac{15}{2} & 12 \end{pmatrix}$$

para y: tomar $X + Y = 3A$

$$\begin{pmatrix} \frac{1}{2} & \frac{25}{2} \\ \frac{15}{2} & 12 \end{pmatrix} + Y = 3 \cdot \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix}$$

$$Y = \begin{pmatrix} -3 & 21 \\ 6 & 18 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & \frac{25}{2} \\ \frac{15}{2} & 12 \end{pmatrix}$$

$$Y = \begin{pmatrix} -\frac{7}{2} & -\frac{17}{2} \\ -\frac{3}{2} & 6 \end{pmatrix}$$

$$6 - A^2 = 2 \cdot A$$

$$\begin{pmatrix} 1 & \frac{1}{x} \\ x & 1 \end{pmatrix}^2 = 2 \cdot \begin{pmatrix} 1 & \frac{1}{x} \\ x & 1 \end{pmatrix}$$

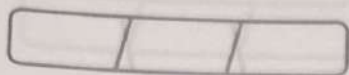
$$\begin{pmatrix} 1 & \frac{1}{x} \\ x & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{x} \\ x & 1 \end{pmatrix} = \begin{pmatrix} 2 & \frac{2}{x} \\ 2x & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 \cdot 1 + \frac{1}{x} \cdot x & \frac{1}{x} \cdot 1 + 1 \cdot \frac{1}{x} \\ x \cdot 1 + 1 \cdot x & x \cdot \frac{1}{x} + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & \frac{2}{x} \\ 2x & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & \frac{2}{x} \\ 2x & 2 \end{pmatrix} = \begin{pmatrix} 2 & \frac{2}{x} \\ 2x & 2 \end{pmatrix}$$

$$A^n = n \cdot A$$





7-a) $A(B+C)$, Sabiendo que:

$$A(B+C) = A \cdot B + A \cdot C$$

$$\therefore A(B+C) = X \cdot Y$$

b) $B^t A^t$, Sabiendo que:

$$B^t A^t = (A \cdot B)^t$$

$$\therefore B^t A^t = (X)^t$$

c) $C^t A^t$, Sabiendo que:

$$C^t \cdot A^t = (A \cdot C)^t$$

$$\therefore C^t \cdot A^t = (Y)^t$$

d) $(ABA) \cdot C$, Sabiendo que

$$(ABA)C = AB \cdot AC$$

$$\therefore (ABA)C = X \cdot Y$$

8-a) $A^t = A$

$$\begin{pmatrix} 4 & 2x-3 \\ x+2 & x+1 \end{pmatrix} = \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}; \text{ Sabiendo que } a_{ij} = a_{ji}$$

$$2x-3 = x+2$$

$$x = 5$$

tendo $x = 5$:

$$A = \begin{pmatrix} 4 & 7 \\ 7 & 4 \end{pmatrix}$$

b) $B^t = -B$

$$\begin{pmatrix} 0 & x & y \\ -4 & 0 & 2z \\ 2 & 1-z & 0 \end{pmatrix} = \begin{pmatrix} 0 & 4 & -2 \\ -x & 0 & -1+z \\ -y & -2z & 0 \end{pmatrix}, \text{ Sabiendo que } b_{ij} = -b_{ji}$$

$$x = 4$$

$$2z = (-1+z)$$

$$y = -2$$

$$z = -1$$

$$\therefore x = 4; y = -2; z = -1$$



9-

$$3 \cdot \begin{pmatrix} x & y \\ z & t \end{pmatrix} = \begin{pmatrix} x & 6 \\ -1 & 2t \end{pmatrix} + \begin{pmatrix} 4 & x+y \\ z+t & 3 \end{pmatrix}$$

$$\begin{pmatrix} 3x & 3y \\ 3z & 3t \end{pmatrix} = \begin{pmatrix} x+4 & 6+(x+y) \\ -1+(z+t) & 2t+3 \end{pmatrix}$$

analisando termo a termo:

$$3x = x + 4$$

$$3y = 6 + x + y$$

$$3z = (-1) + z + t$$

$$3t = 2t + 3$$

$$2x = 4$$

$$2y = 6 + x$$

$$2z = (-1) + 3$$

$$t = 3$$

$$x = 2$$

$$2y = 6 + 2$$

$$2z = 2$$

$$2y = 8$$

$$z = 1$$

$$y = 4$$

$$\therefore x = 2; y = 4; z = 1; t = 3$$

10-a) $R(\theta)$ é ortogonal se $R(\theta) \cdot (R(\theta))^T = I_n$

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \cdot \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} \cos \theta \cdot \cos \theta + \sin \theta \cdot \sin \theta & \cos \theta \cdot (-\sin \theta) + \sin \theta \cdot \cos \theta \\ -\sin \theta \cdot \cos \theta + \cos \theta \cdot \sin \theta & -\sin \theta \cdot (-\sin \theta) + \cos \theta \cdot \cos \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & \sin^2 \theta + \cos^2 \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \therefore \text{A matriz } R(\theta) \text{ é ortogonal}$$

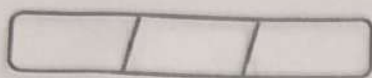
$$b) \begin{pmatrix} 1 & 0 & x \\ 0 & \frac{1}{\sqrt{2}} & y \\ 0 & \frac{1}{\sqrt{2}} & z \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ x & y & z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 + x \cdot x & 1 \cdot 0 + 0 \cdot \frac{1}{\sqrt{2}} + x \cdot y & 1 \cdot 0 + 0 \cdot \frac{1}{\sqrt{2}} + x \cdot z \\ 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 + y \cdot x & 0 \cdot 0 + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + y \cdot y & 0 \cdot 0 + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + y \cdot z \\ 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 + z \cdot x & 0 \cdot 0 + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + z \cdot y & 0 \cdot 0 + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + z \cdot z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1+x^2 & xy & xz \\ xy & \frac{1}{2}+y^2 & \frac{1}{2}+yz \\ xz & \frac{1}{2}+yz & \frac{1}{2}+z^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



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$$1 + x^2 = 1$$

$$x^2 = 0$$

$$x = 0$$

$$\frac{1}{2} + y^2 = 1$$

$$1 + 2y^2 = 2$$

$$2y^2 = 1$$

$$y^2 = \frac{1}{2}$$

$$y = \pm \frac{1}{\sqrt{2}}$$

$$y = \pm \frac{\sqrt{2}}{2}$$

$$\frac{1}{2} + z^2 = 1$$

$$z = \pm \frac{1}{\sqrt{2}}$$

$$z = \pm \frac{\sqrt{2}}{2}$$

$$\frac{1}{2} + y + z = 0$$

$$\text{Se } y = +\frac{\sqrt{2}}{2}, \text{ ent\~{a}o}$$

$$\frac{1}{2} + \frac{\sqrt{2}}{2} \cdot z = 0$$

$$1 + \sqrt{2} \cdot z = 0$$

$$\sqrt{2} \cdot z = -1$$

$$z = -\frac{1}{\sqrt{2}} \Rightarrow -\frac{\sqrt{2}}{2}$$

$$\text{ou se } y = -\frac{\sqrt{2}}{2}, \text{ ent\~{a}o}$$

$$z = +\frac{\sqrt{2}}{2}$$

